

Apache Kafka — Core Concepts

1. Message

A **message** is a piece of data or an event that is sent to Kafka.

It represents information that you want Kafka to store and make available for processing.

Simple Definition

A message is a record of data or an event sent to Kafka.

Example

```
"Ahmed placed an order for a laptop"
```

2. Producer

A **producer** is an application or system that sends messages to Kafka.

The producer decides which **topic** the message should be sent to.

Simple Definition

A producer is an application that sends messages to Kafka topics.

Example

```
Order Application
  ↓
  Producer
  ↓
orders topic
```

3. Topic

A **topic** is a named stream or category used to organize messages in Kafka.

You can think of a topic like a **folder for a particular type of data**.

For example:

```
orders
payments
customers
transactions
```

A topic is divided into **partitions**, and Kafka stores messages within those partitions.

Simple Definition

A topic is a named stream or category used to organize and store messages in Kafka.

Example

```
Topic: orders

Message 1 → Ahmed placed an order
Message 2 → Sara placed an order
Message 3 → Ali placed an order
```

4. Consumer

A **consumer** is an application or system that reads messages from Kafka.

The consumer can then process the messages or send the data to another system.

Simple Definition

A consumer is an application that reads and processes messages from Kafka topics.

Important

A consumer reading a message does **not normally remove the message from Kafka**.

Kafka keeps messages according to its retention configuration.

Example

```
orders topic
  ↓
  Consumer
  ↓
Process order
```

5. Broker

A **broker** is a Kafka server.

A Kafka cluster can contain multiple brokers, and each broker can store partitions from different topics.

Simple Definition

A broker is a Kafka server that stores and serves messages.

Main responsibilities

A broker can:

- Receive messages from producers
- Store messages
- Serve messages to consumers
- Manage partitions
- Participate in replication

Example

Kafka Cluster

Broker 1

Broker 2

Broker 3

6. Partition

A **partition** is an ordered sequence of messages inside a Kafka topic.

A topic can contain one or many partitions.

Partitions allow Kafka to **scale and process data in parallel**.

Simple Definition

A partition is an ordered sequence of messages within a topic that enables scalability and parallel processing.

Important

Messages within the **same partition are ordered**.

For example:

Partition 0

Message A
Message B
Message C
Message D

Kafka preserves this order:

A → B → C → D

However, you should not assume ordering across different partitions.

Example

Topic: orders

├─ Partition 0
├─ Partition 1
└─ Partition 2

7. Offset

An **offset** is a sequential position number assigned to a message within a partition.

The offset identifies the position of a message in that partition.

Simple Definition

An offset is the sequential position of a message within a Kafka partition.

Example

Partition 0

Offset 0 → Message A
Offset 1 → Message B
Offset 2 → Message C
Offset 3 → Message D

The offset allows a consumer to keep track of its position while reading messages.

For example:

Consumer processed:

Offset 0

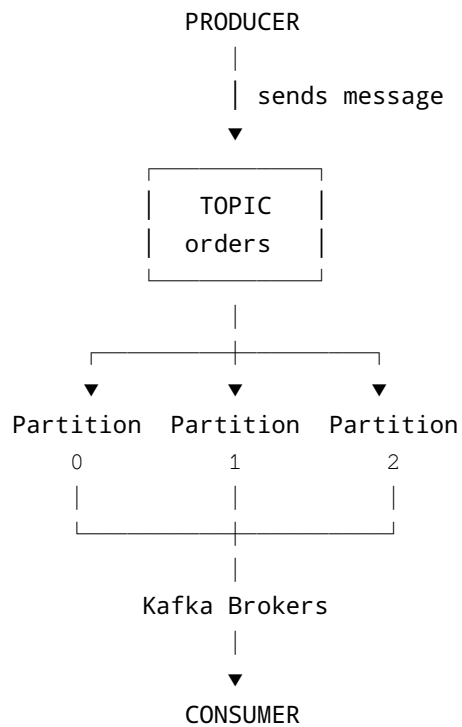
Offset 1

Offset 2

Next message:

Offset 3

How the Concepts Work Together



Inside a Partition

Partition 0

Offset 0 → Message A

Offset 1 → Message B

Offset 2 → Message C

Offset 3 → Message D

Quick Definitions

Message:

A record of data or an event sent to Kafka.

Producer:

An application that sends messages to Kafka topics.

Topic:

A named stream or category used to organize messages.

Consumer:

An application that reads and processes messages from Kafka topics.

Broker:

A Kafka server that stores and serves messages.

Partition:

An ordered sequence of messages within a topic that enables scalability and parallel processing.

Offset:

A sequential position number assigned to a message within a partition.

Mental Model

Producer



Message



Topic



Partitions



Broker



Consumer

Remember

Producer → SENDS

Topic → ORGANIZES

Partition → DIVIDES

Broker → STORES / SERVES

Consumer → READS
Offset → TRACKS POSITION
Message → DATA / EVENT